

Homework Assignment 4

Chloe Lo Papa

2026-05-24

Question 1

Homework #4

① set up the design matrix X and B vector for each of the following regression models (assume $i=1,2,3,4$)

a) $Y_i = B_0 + B_1 X_{i2} + B_2 X_{i2} X_{i1} + \epsilon_i$

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \end{bmatrix} \quad X = \begin{bmatrix} 1 & X_{11} & X_{11}X_{12} \\ 1 & X_{21} & X_{21}X_{22} \\ 1 & X_{31} & X_{31}X_{32} \\ 1 & X_{41} & X_{41}X_{42} \end{bmatrix} \quad B = \begin{bmatrix} B_0 \\ B_1 \\ B_2 \end{bmatrix}$$

b) $\log Y_i = B_0 + B_1 X_{i2} + B_2 X_{i2} + \epsilon_i$

$$Y = \begin{bmatrix} \log Y_1 \\ \log Y_2 \\ \log Y_3 \\ \log Y_4 \end{bmatrix} \quad X = \begin{bmatrix} 1 & X_{11} & X_{12} \\ 1 & X_{21} & X_{22} \\ 1 & X_{31} & X_{32} \\ 1 & X_{41} & X_{42} \end{bmatrix} \quad B = \begin{bmatrix} B_0 \\ B_1 \\ B_2 \end{bmatrix}$$

c) $Y_i = B_0 X_{i1} + B_1 X_{i2} + B_2 X_{i1}^2 + \epsilon_i$

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ Y_4 \end{bmatrix} \quad X = \begin{bmatrix} X_{11} & X_{12} & X_{11}^2 \\ X_{21} & X_{22} & X_{21}^2 \\ X_{31} & X_{32} & X_{31}^2 \\ X_{41} & X_{42} & X_{41}^2 \end{bmatrix} \quad B = \begin{bmatrix} B_0 \\ B_1 \\ B_2 \end{bmatrix}$$

d) $\sqrt{Y_i} = B_0 + B_1 X_{i2} + B_2 \log_{10} X_{i2} + \epsilon_i$

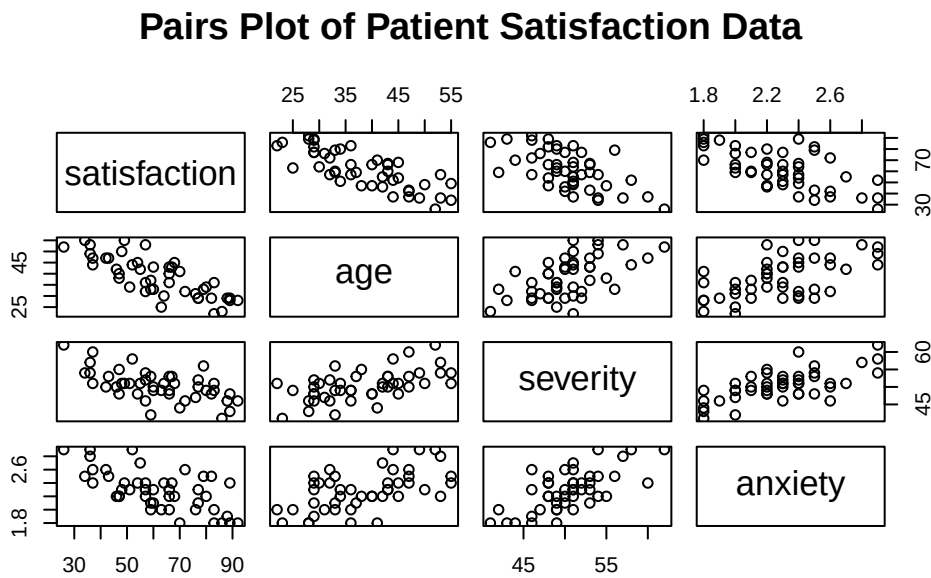
$$Y = \begin{bmatrix} \sqrt{Y_1} \\ \sqrt{Y_2} \\ \sqrt{Y_3} \\ \sqrt{Y_4} \end{bmatrix} \quad X = \begin{bmatrix} 1 & X_{11} & \log_{10}(X_{12}) \\ 1 & X_{21} & \log_{10}(X_{22}) \\ 1 & X_{31} & \log_{10}(X_{32}) \\ 1 & X_{41} & \log_{10}(X_{42}) \end{bmatrix} \quad B = \begin{bmatrix} B_0 \\ B_1 \\ B_2 \end{bmatrix}$$

Figure 1: Handwritten Solutions for Q1

Question 2 A hospital administrator wished to study the relation between patient satisfaction and patient's age (in years), severity of illness (an index), and anxiety level (an index). The administrator randomly selected 46 patients and collected the data. Larger values of patient satisfaction, illness, and anxiety are, respectively, associated with more satisfaction, increased severity of illness, and more anxiety.

a) Prepare pairs plot for all variables. Do these plots reveal any noteworthy features?

```
1 patients <- read.table("patient_satisfaction.txt", header=TRUE)
2 pairs(patients, main = "Pairs Plot of Patient Satisfaction Data")
```



Satisfaction has a strong negative correlation with the factors age, severity and anxiety. Age and severity have a positive correlation, same with age and anxiety although it is less strong. Severity and anxiety have a strong positive correlation. All of these relationships look relatively linear.

b) Fit a regression model with all three predictor variables to the data and state the estimated regression function. Interpret each estimated parameter.

```
1 fit <- lm(satisfaction ~ age + severity + anxiety, data=patients)
2 summary(fit)
```

Call:

```
lm(formula = satisfaction ~ age + severity + anxiety, data = patients)
```

Residuals:

Min	1Q	Median	3Q	Max
-18.3524	-6.4230	0.5196	8.3715	17.1601

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	158.4913	18.1259	8.744	5.26e-11 ***
age	-1.1416	0.2148	-5.315	3.81e-06 ***
severity	-0.4420	0.4920	-0.898	0.3741
anxiety	-13.4702	7.0997	-1.897	0.0647 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.06 on 42 degrees of freedom

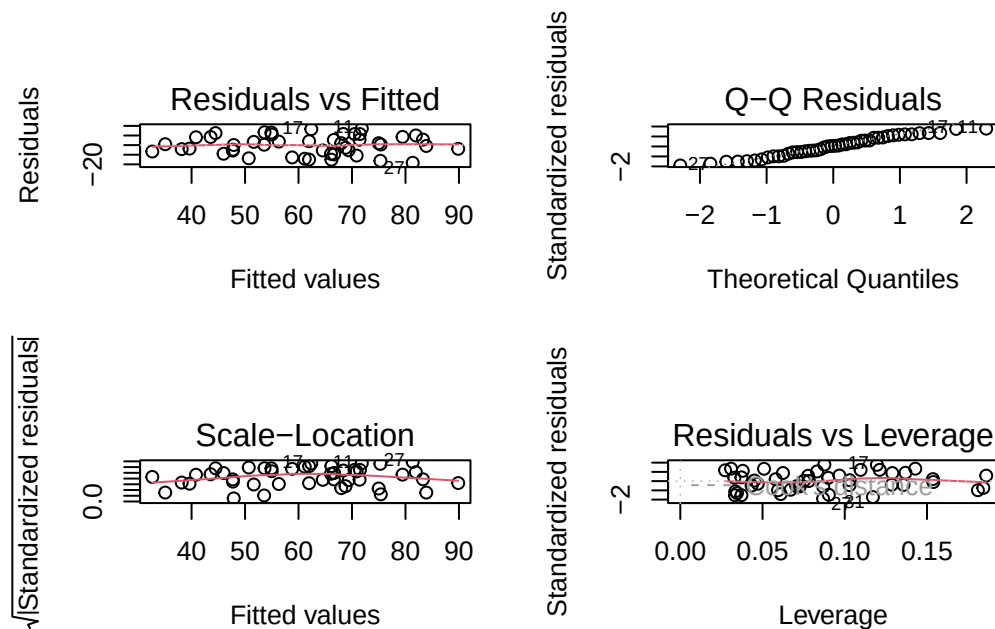
Multiple R-squared: 0.6822, Adjusted R-squared: 0.6595

F-statistic: 30.05 on 3 and 42 DF, p-value: 1.542e-10

$\hat{Y} = 158.4913 - 1.1416(\text{age}) - 0.4420(\text{severity}) - 13.4702(\text{anxiety})$ The intercept represents the predicted patient satisfaction when age severity and anxiety =0 Holding severity and anxiety constant a one year increase in age is associated with about a 1.1416 point decrease in satisfaction Holding age and anxiety constant a one unit increase in severity is associated with a 0.4420 point decrease in satisfaction Holding age and severity constant a one unit increase in anxiety is associated with about a 13.4702 point decrease in satisfaction.

c) Construct diagnostic plots and comment on your findings

```
1 fit <- lm(satisfaction ~ age + severity + anxiety, data=patients)
2 par(mfrow=c(2,2))
3 plot(fit)
```



The residuals vs fitted plot looks randomly scattered around zero so it seems reasonable to assume linearity and constant variance The QQ residuals plot tells us residuals are approximately normally distributed because they closely follow the diagonal line Scale Location the has the trend line as roughly horizontal making me think there is constant variance Residuals vs leverage line, there doesn't seem to be large outliers

d) Report R2 and comment.

```
1 summary(fit)
```

Call:

```
lm(formula = satisfaction ~ age + severity + anxiety, data = patients)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-18.3524	-6.4230	0.5196	8.3715	17.1601

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	158.4913	18.1259	8.744	5.26e-11 ***
age	-1.1416	0.2148	-5.315	3.81e-06 ***
severity	-0.4420	0.4920	-0.898	0.3741
anxiety	-13.4702	7.0997	-1.897	0.0647 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.06 on 42 degrees of freedom

Multiple R-squared: 0.6822, Adjusted R-squared: 0.6595

F-statistic: 30.05 on 3 and 42 DF, p-value: 1.542e-10

The .68 R^2 means that 68% of the variability in patient satisfaction can be explained by age, severity and anxiety

- e) Assuming the fitted model is appropriate, state estimates of all parameters with 95% confidence intervals.

```
1 confint(fit, level = 0.95)
```

	2.5 %	97.5 %
(Intercept)	121.911727	195.0707761
age	-1.575093	-0.7081303
severity	-1.434831	0.5508228
anxiety	-27.797859	0.8575324

0 (Intercept) 158.4913 95% Interval (121.9117, 195.0708) 1(Age) -1.1416 95% Interval (-1.5751, -0.7081)
2(Severity) -0.4420 95% Interval (-1.4348, 0.5508) 3(Anxiety) -13.4702 95% Interval (-27.7979, 0.8575)

- f) Obtain an estimate of the mean satisfaction with 90% confidence interval when age = 35, severity of illness = 45, and anxiety level = 2.2

```
1 newpatient <- data.frame(age = 35, severity = 45, anxiety = 2.2)
2 predict(fit, newdata = newpatient, interval = "confidence", level = 0.90)
```

	fit	lwr	upr
1	69.01029	64.52854	73.49204

$Y(\text{hat}) = 158.4913 - 1.1416(35) - 0.4420(45) - 13.4702(2.2)$ $Y(\text{hat}) = 69.01$

- g) Obtain a 90% prediction interval for a new patient's satisfaction when age = 35, severity of illness = 45, and anxiety level = 2.2

The confidence interval is (64.52, 73.49)

- h) The hospital administrator wishes to determine the best subset of predictor variables for predicting patient satisfaction. Indicate which subset of predictor variables you would recommend as best for predicting patient satisfaction according to the stepwise procedure using AIC.

```
1 step<fit, direction = "both">
```

Start: AIC=216.18

satisfaction ~ age + severity + anxiety

	Df	Sum of Sq	RSS	AIC
- severity	1	81.66	4330.5	215.06
<none>			4248.8	216.19
- anxiety	1	364.16	4613.0	217.97
- age	1	2857.55	7106.4	237.84

Step: AIC=215.06

satisfaction ~ age + anxiety

	Df	Sum of Sq	RSS	AIC
<none>			4330.5	215.06
+ severity	1	81.7	4248.8	216.19
- anxiety	1	763.4	5093.9	220.53
- age	1	3483.9	7814.4	240.21

Call:

```
lm(formula = satisfaction ~ age + anxiety, data = patients)
```

Coefficients:

(Intercept)	age	anxiety
145.94	-1.20	-16.74

the variable severity is not statistically significant, while age is highly significant and anxiety is marginally significant. I would include a model with only age and anxiety so that it is simpler $\hat{Y} = 0 + 1(\text{age}) + 2(\text{anxiety})$

- i) Does the stepwise procedure in (h) identify the same best subset by exhaustive search? Does this always happen? Exhaustive search guarantees finding the model with the minimum AIC because it evaluates every possible subset.

```
1 models <- list(
2   age = lm(satisfaction ~ age, data=patients),
3   severity = lm(satisfaction ~ severity, data=patients),
4   anxiety = lm(satisfaction ~ anxiety, data=patients),
5   age_severity = lm(satisfaction ~ age + severity, data=patients),
6   age_anxiety = lm(satisfaction ~ age + anxiety, data=patients),
7   severity_anxiety = lm(satisfaction ~ severity + anxiety, data=patients),
8   full = lm(satisfaction ~ age + severity + anxiety, data=patients)
9 )
10
11 sapply(models, AIC)
```

age	severity	anxiety	age_severity
353.0717	376.6735	372.7561	350.5100
age_anxiety	severity_anxiety	full	
347.6030	370.3874	348.7273	

Age and Anxiety still chosen in this case

- j) Would the stepwise procedure have any advantages as a screening procedure over the exhaustive search?

Stepwise regression evaluates far fewer models than exhaustive search making it more practical for large searches

1 Question 3

- a) Provide comment on the preliminary plots. Rental vs. footage There is a fairly strong positive relationship. Larger commercial properties tend to have higher rental rates.

Rental vs. age There is a noticeable negative relationship. Older buildings tend to have lower rental rates

Rental vs. expenses There appears to be a positive association between expenses and rental rates. Properties with higher operating expenses tend to charge higher rents.

Rental vs. vacancy There is little obvious relationship between rental rate and vacancy percentage. The points appear widely scattered.

The plots don't look very obviously linear

- b) For the saved R object fit, write down the model that is being fitted, including assumptions. $\text{rental}_i = 0 + 1(\text{age}_i) + 2(\text{expenses}_i) + 3(\text{vacancy}_i) + 4(\text{footage}_i) + \text{error}_i$ Assumptions: the mean rental rate is linearly related to the predictors, the error terms are independent, homoscedasticity, normality of errors, mean zero for errors
- c) Explain the purpose of the statement `> (fit1 <- stepAIC(lm(rental~1), scope=formula(fit), direction='forward'))` Provide details about the criterion and the option direction. Discuss the outcome of each step and the final model

performs forward stepwise variable selection for multiple linear regression using the AIC The stepAIC() function then sequentially adds predictor variables from the full model, begins with the intercept-only model The selected model is: $\text{rental} = 0 + 1(\text{footage}) + 2(\text{age}) + 3(\text{expenses}) +$

Starts with B0 and adds footage because it has the lowest increase on AIC, then add age because it improves AIC, then add expenses because it produces smallest AIC, vacancy would increase AIC so it should not be added to the model.

- d) Suppose the final model fit1 is adequate. State your conclusions. conclude that the regression model is highly significant overall. At least one predictor variable is associated with rental rates because the p value is so small. approximately 58% of the variability in rental rates is explained by the predictors in the model. The model provides a moderate fit to the data.
- e) Comment on the residual plots. Are you satisfied with the model fit1, and explain why.

Residuals vs Fitted The residuals are scattered fairly randomly around zero with no pattern and The spread of residuals appears relatively constant across fitted values with no severe outliers The assumptions of linearity and constant variance appear reasonably satisfied

QQ plot Most points lie close to the reference line. There are mild deviations in the tails but not too extreme The residuals appear approximately normally distributed.

the model fit1 appears satisfactory because the assumptions are appropriate based on the two plots